

Automation System Design

Smart Home Lighting Control System

Domain:	Robotics & Automation
Task:	Task 3 — Automation System Design
System Type:	Smart Home Lighting Automation
Organization:	CodeAlpha

1. System Overview

The Smart Home Lighting Control System is an IoT-enabled automation solution designed to optimize energy consumption and enhance user comfort by automatically managing indoor lighting based on occupancy, ambient light levels, and user preferences. The system integrates passive infrared (PIR) motion sensors, light-dependent sensors, a microcontroller, and a mobile/cloud interface to deliver intelligent, responsive lighting across multiple room zones.

The primary design objectives are: (1) automatic illumination when a person enters a room; (2) adaptive brightness based on natural light availability; (3) automatic shut-off after a configurable idle period; (4) manual override capability via touch panel or mobile app; and (5) energy usage monitoring and reporting. Studies show that automated lighting systems reduce residential electricity consumption by 30–50% compared to manual switching (ENERGY STAR, 2022).

2. System Flowchart

The flowchart below illustrates the complete control logic of the smart lighting system, from system initialization through sensor-based decision-making to energy-saving idle mode.

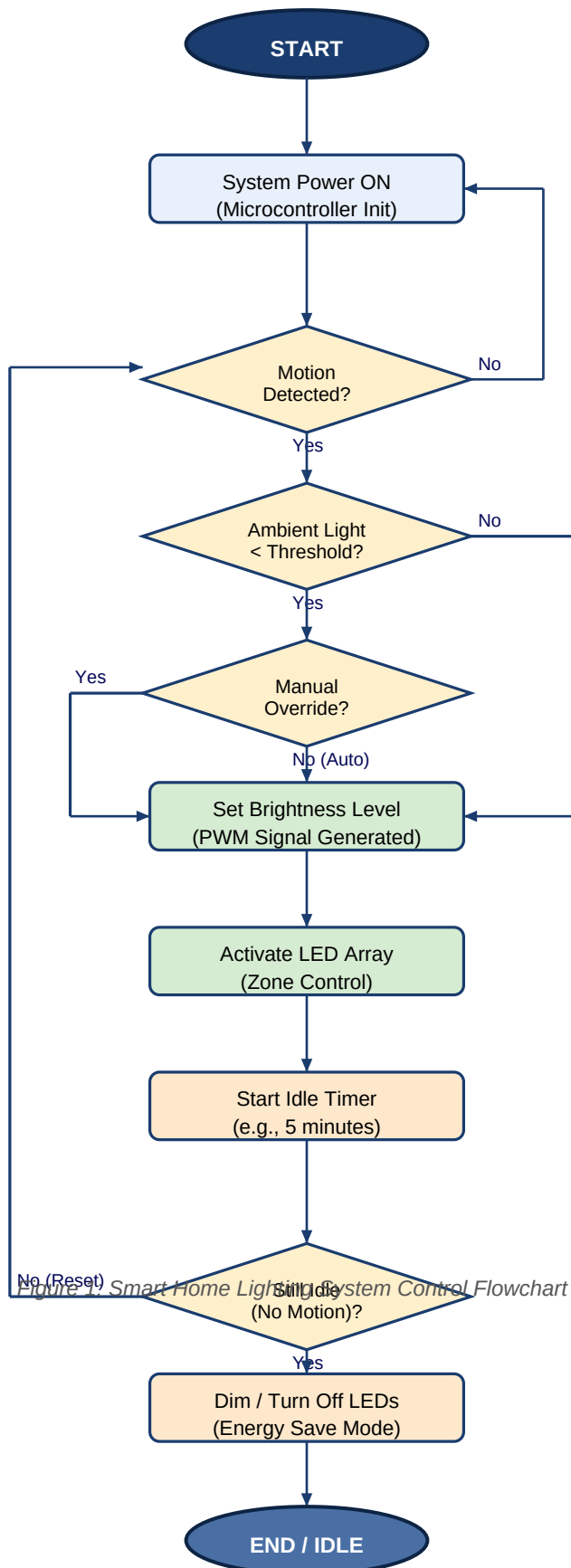


Figure 1. Smart Home Lighting System Control Flowchart

3. Component Specifications

The table below lists all hardware components used in the system design, along with their specifications and primary functions within the control architecture.

Component	Specification	Function
Microcontroller	Arduino Mega 2560 / ESP32	Central processing & PWM control
PIR Motion Sensor	HC-SR501 (Range: 7m, 120°)	Detects human presence in room
LDR / Light Sensor	TEMT6000 (0-1000 lux)	Measures ambient illumination level
LED Lighting Array	12V RGB LED Strip (5A/ch)	Dimmable room lighting output
MOSFET Driver	IRLZ44N (55V, 47A)	High-current LED switching
Wi-Fi Module	ESP8266 / ESP32 built-in	Remote control & IoT connectivity
Power Supply	12V 10A SMPS	System power regulation
Touch/Button Panel	Capacitive TTP223B	Manual override input
Relay Module	5V 4-channel relay	Zone isolation & safety cutoff

Table 1: Hardware Components and Specifications

4. Circuit and System Architecture Description

4.1 Sensor Input Layer

The HC-SR501 PIR sensor detects infrared radiation emitted by human bodies, generating a HIGH digital signal on detection that is read by the microcontroller's GPIO pin. The sensitivity and time delay are adjustable via onboard potentiometers. The TEMT6000 ambient light sensor produces an analog voltage proportional to illuminance (0–3.3V corresponding to 0–1000 lux), read through the microcontroller's ADC channel. A configurable threshold (default: 300 lux) determines when artificial lighting is required.

4.2 Processing Layer

The ESP32 microcontroller forms the central processing unit. It runs a real-time control loop sampling sensor inputs every 500ms. When motion is detected AND ambient light falls below the threshold, the microcontroller calculates the required PWM duty cycle inversely proportional to ambient light level—brighter natural light results in lower LED intensity, maintaining a consistent target illuminance of 400–500 lux at working height (ISO 8995-1:2002 office standard). The LEDC peripheral of the ESP32 generates PWM signals at 5 kHz to eliminate visible flicker.

4.3 Output and Power Layer

PWM signals from the ESP32 drive the gates of IRLZ44N N-channel MOSFETs, which switch the 12V LED strip current. Each MOSFET controls an independent zone, allowing differential brightness

across rooms. A 12V 10A switch-mode power supply provides regulated voltage, with the 5V rail for the ESP32 derived via an LM2596 buck converter. The 4-channel relay module provides galvanic isolation for safety and enables complete power cutoff for each zone.

4.4 Connectivity and User Interface

The ESP32's built-in Wi-Fi connects to the home router, exposing an HTTP API that interfaces with a custom mobile application (or standard platforms such as Home Assistant via MQTT protocol). Users can view real-time energy consumption, adjust brightness schedules, set scene presets, and override automatic behavior. A local capacitive touch panel provides immediate manual control without reliance on network connectivity.

5. Operating Modes Summary

Mode	Trigger Condition	LED State	Notes
AUTO-ON	Motion + Low Ambient Light	ON (Adaptive PWM)	Primary operating mode
STANDBY	Motion + Sufficient Natural Light	OFF	Energy saving during daytime
IDLE-DIM	No Motion > 3 minutes	10% brightness	Grace period warning
IDLE-OFF	No Motion > 5 minutes	OFF	Full energy saving
MANUAL	Touch/App override	User-defined	Bypasses all automation
NIGHT	Scheduled 11PM–6AM	5% dim (if motion)	Sleep-friendly mode
EMERGENCY	Alarm signal HIGH	100% brightness	Safety override

Table 2: System Operating Modes

6. Expected Benefits and Performance Metrics

The smart lighting system is designed to reduce household lighting energy consumption by an estimated 35–45% compared to conventional manual switching, based on occupancy patterns typical of a 3-bedroom home (IEA, 2022). The adaptive dimming feature can further reduce energy use by 20% during daylight hours. System response time from motion detection to full illumination is under 200ms, ensuring a seamless user experience. All components are rated for a minimum operational lifespan of 50,000 hours for LEDs and 10+ years for the electronic control system.

From a cost perspective, the complete system bill of materials is estimated at USD 45–80 for a single room zone, with payback periods of 18–24 months based on average electricity tariffs. The modular architecture supports straightforward expansion to additional zones without redesigning the core control logic.

7. Conclusion

This document has presented the complete design of a Smart Home Lighting Automation System, encompassing the system flowchart, hardware component specifications, circuit architecture, operating modes, and expected performance outcomes. The design demonstrates the practical application of sensor integration, microcontroller-based control logic, and IoT connectivity to create an intelligent, energy-efficient lighting solution. The system is scalable, cost-effective, and aligned with modern smart home standards, making it a viable candidate for residential and small commercial deployment.

References

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